Executive Business Report
End-to-End Sales Forecasting & Demand Intelligence System
Prepared For:
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LiveLink:https://salesforecastingtanvimarkadgitfcb3hg7p2vyjapskcek2u5.streamlit.app/

Executive Summary

This project analyzes four years of retail sales data to understand customer demand, identify unusual sales patterns, forecast future sales, and recommend better inventory strategies. Multiple forecasting techniques were evaluated, including SARIMA, Prophet, and XGBoost. Among these, **XGBoost produced the highest forecasting accuracy**, making it the preferred model for predicting future sales. The system also identifies abnormal sales events and groups products based on demand patterns to support inventory planning. These insights help reduce inventory costs, improve product availability, and support better business decisions.

Key Findings from Data Analysis

1. Product Category Performance

The **Technology** category generated the highest total revenue, indicating consistently strong customer demand. Furniture and Office Supplies contributed steadily but at comparatively lower levels.

2. Regional Sales Analysis

Sales remained relatively stable across all regions, with the **West region showing the most consistent sales growth** over the four-year period.

3. Shipping Performance

The average shipping time between Order Date and Ship Date was approximately **4 days**, with only small differences observed across regions.

4. Seasonality

Sales consistently increased during **November and December**, indicating strong seasonal demand driven by year-end shopping and promotional events.

Forecasting Results

Three forecasting models were developed and evaluated.

Model	**MAE**	**RMSE**	**MAPE**
SARIMA	18031.40	19009.18	0.1897
Prophet	20250.79	22318.41	0.2186
XGBoost	**14443.46**	**17069.09**	**0.1445**

The evaluation shows that **XGBoost achieved the lowest prediction error**, making it the most accurate forecasting model.

Three-Month Sales Forecast

The forecasting models indicate that sales are expected to remain **stable with a gradual upward trend over the next three months**. The confidence intervals suggest moderate uncertainty, but overall demand is expected to remain healthy. This forecast supports proactive inventory planning and purchasing decisions.

Anomaly Detection

The anomaly detection process identified several unusual sales observations.

Top Three Likely Causes

1. Exceptionally high sales during major promotional campaigns and festive seasons.
2. Unexpected drops in sales caused by reduced customer demand or temporary supply shortages.
3. Large bulk purchase orders from corporate customers creating temporary spikes in monthly sales.

These anomalies should be monitored separately because they may not represent normal business behaviour.

Product Demand Segmentation

Products were grouped into four demand clusters using K-Means clustering.

Cluster 0 – High Demand

- Fast-moving products
- Maintain higher inventory levels
- Frequent replenishment recommended

Cluster 1 – Moderate Demand

- Stable sales
- Maintain balanced inventory
- Monthly inventory review recommended

Cluster 2 – Low Demand

- Slow-moving products
- Keep minimum stock
- Consider promotional campaigns to improve sales

Cluster 3 – Seasonal Demand

- Demand changes throughout the year
- Increase stock before seasonal peaks
- Reduce inventory during off-season periods

Business Recommendations

Recommendation 1

Increase inventory levels for high-demand Technology products before peak sales months. Historical sales data shows consistently higher revenue from this category.

Recommendation 2

Prepare additional inventory during November and December because historical sales demonstrate strong seasonal demand during these months.

Recommendation 3

Adopt the XGBoost forecasting model for monthly inventory planning because it achieved the **lowest forecasting error (MAE = 14,443.46, RMSE = 17,069.09, MAPE = 14.45%)**, providing the most reliable predictions.

Business Risk / Limitation

The forecasting system relies entirely on historical sales data. Unexpected events such as economic changes, supplier disruptions, policy changes, or sudden shifts in customer behaviour may reduce forecasting accuracy. Therefore, forecasts should be reviewed regularly and updated as new sales data becomes available.

Conclusion

The End-to-End Sales Forecasting & Demand Intelligence System provides valuable insights into sales performance, customer demand, inventory optimization, and future sales trends. By combining exploratory data analysis, anomaly detection, machine learning forecasting, and demand segmentation, the system enables better strategic planning and inventory management. Based on the evaluation results, **XGBoost is recommended as the primary forecasting model**, helping the business improve forecast accuracy, reduce inventory costs, and maintain high product availability.